Causal Inference

Estimating the Causal Effect of Delivery Delay on Customer Ratings in E-commerce

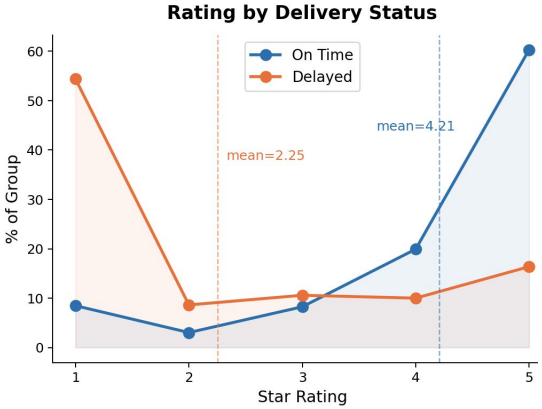
Runyan Xin, SeongJun Kwon

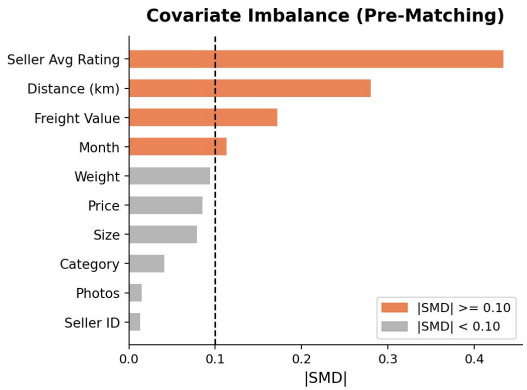
What is the Dataset? - EDA

Dataset Overview

Variable	Value
Orders	114,841
Delayed	7,368 (6.4%)
On Time	107,473 (93.6%)
Treatment T	is_delivery_late
Outcome Y	Star Rating (1-5)
Covariates	10
Period	2017-2018
Source	Olist (Brazil)

Covariates: distance_km, freight_value, price, product_weight_kg, product_size, month, rainfall, seller_avg_rating, product_category_name_encoded, no_photos, seller_id_encoded





OLS Regression: Effect of Delivery Delay on Rating

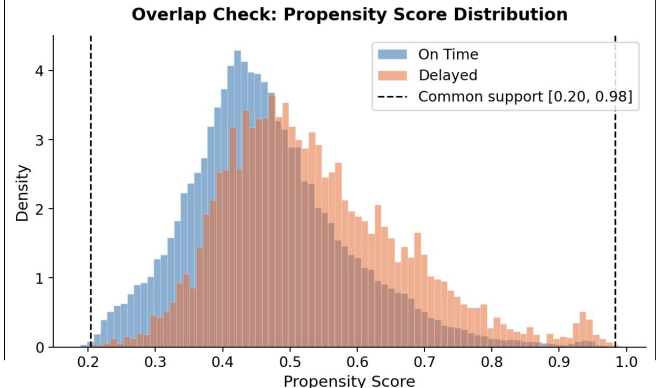
	Bivariate OLS	OLS + Controls
Coefficient	-1.952	-1.744
Std. Error	0.019	0.019
95% CI	[-1.989, -1.916]	[-1.781, -1.708]
p-value	< 0.001	< 0.001
R-squared	0.126	0.218
N	114,841	114,841
Controls	No	Yes

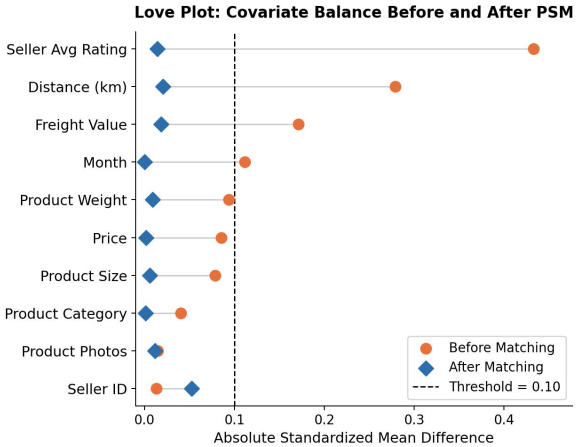
Propensity Score Matching

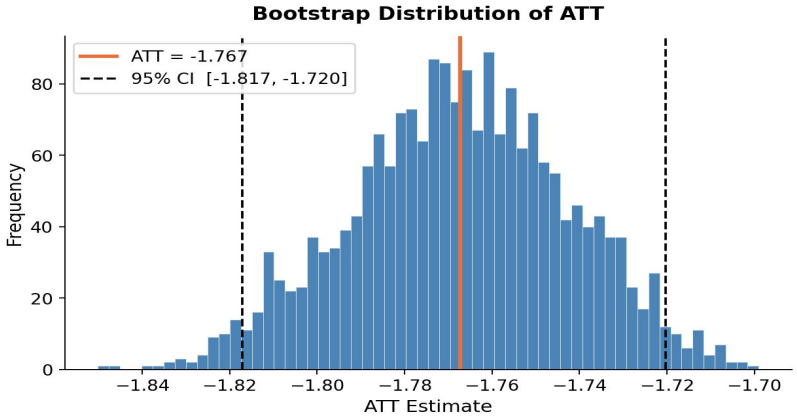
Workflow:

- Covariates X, Treatment $T = \texttt{is_delivery_late}$, Outcome $Y = \text{Rating}$
- Step 1: estimate $e(X) = P(T = 1 \mid X)$
- Step 2: trim to common support
- Step 3: 1:1 nearest-neighbor matching with caliper
- Step 4: check balance $|\text{SMD}| < 0.10$
- Step 5: $\widehat{\text{ATT}} = \frac{1}{n_1} \sum_{i:T_i=1} (Y_i - Y_{j(i)})$

overlap check







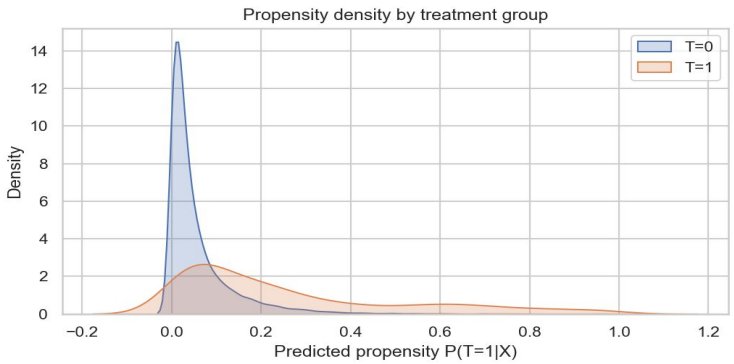
1. Matching successfully eliminated covariate imbalance
2. Late delivery causally reduces ratings by 1.77 stars

Heterogeneous treatment effects(DML + Causal Forest)

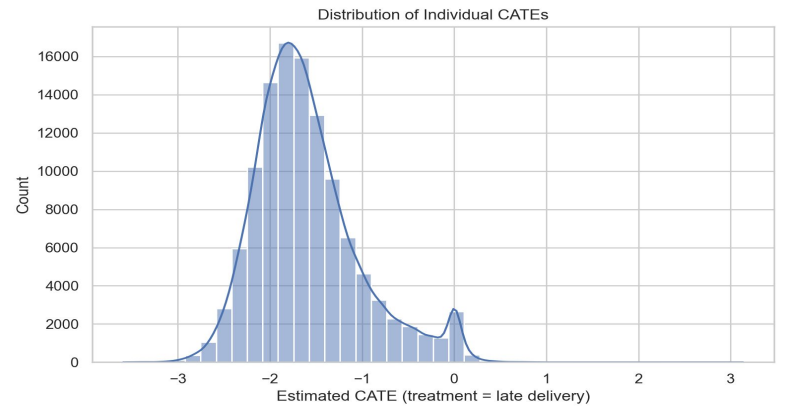
Workflow:

- Covariates X
- Treatment T = is_delivery_late
- Outcome Y = Rating
- Step 1: estimate nuisance functions
  - $e(X) = P(T = 1 \mid X)$
  - $m(X) = E[Y \mid X]$
- Step 2: orthogonalization / residualization
- Step 3: causal forest estimates CATE $\tau(X)$

overlap check

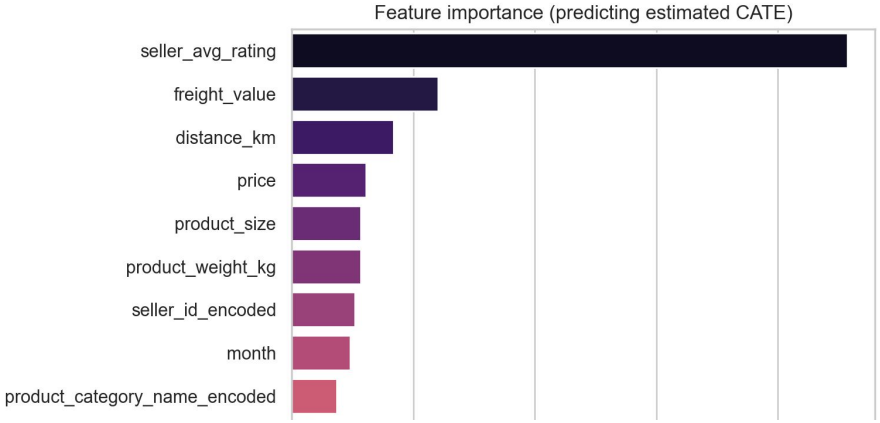


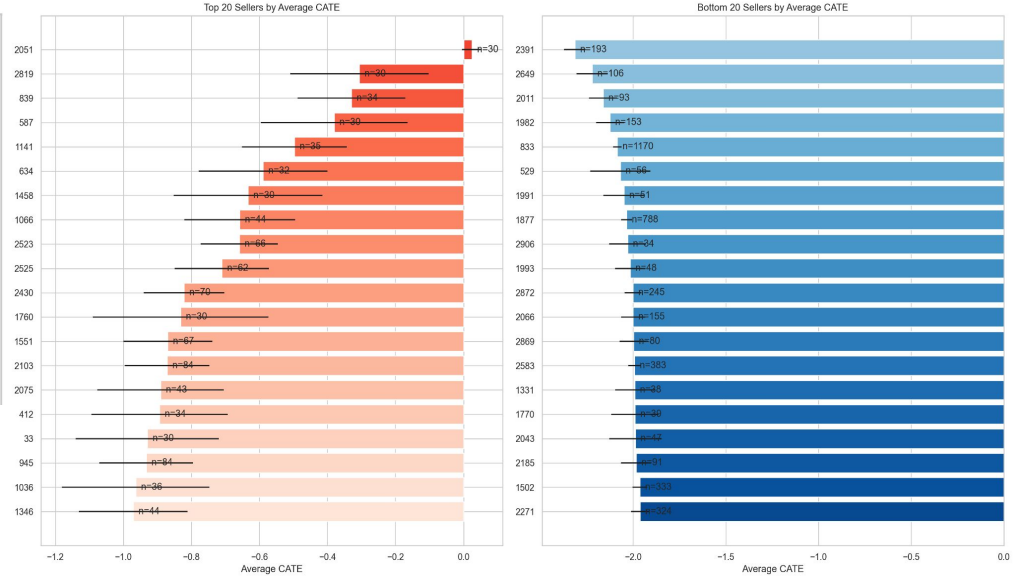
Distribution of individual CATEs

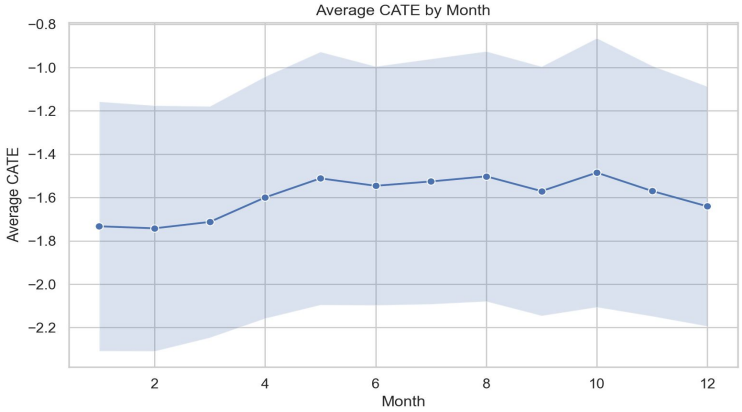


1. DML + Causal Forest estimates individual CATEs
2. Most estimated CATEs are negative, indicating that late delivery tends to reduce ratings
3. Overlap is acceptable for estimation

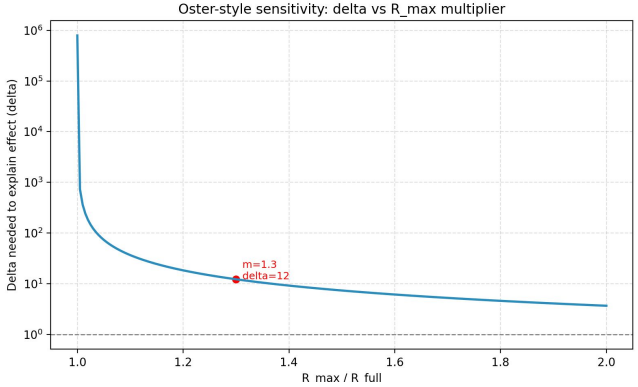
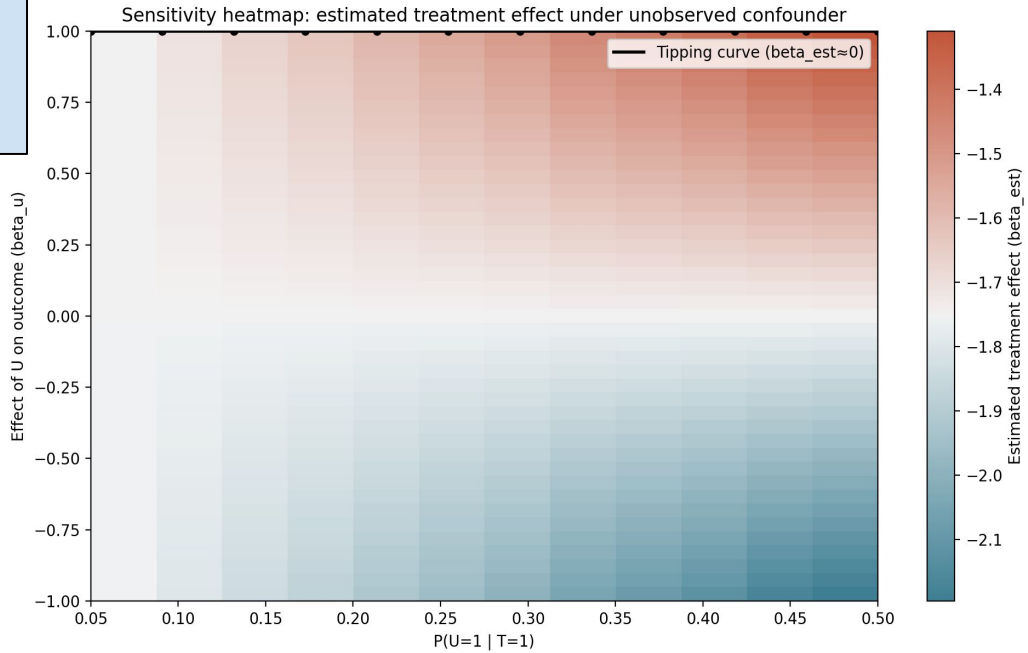
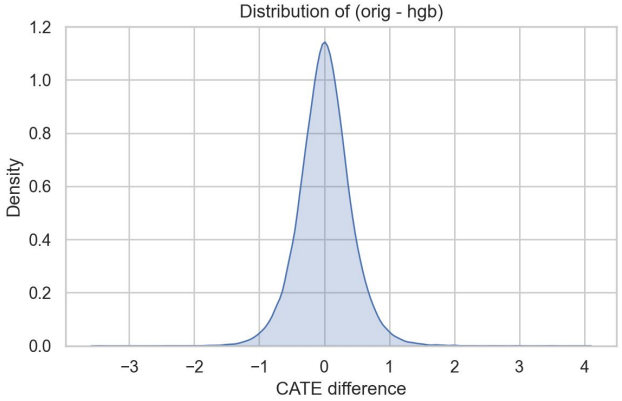
Estimated CATE and key sources of heterogeneity







Feature importance: seller quality and shipping-related factors are the main drivers of heterogeneity.
By month: average CATE remains negative across all months.
By seller: treatment effects vary substantially across sellers.

Robustness and sensitivity analysis

1. model-specification sensitivity analysis
   - Distribution of (orig - hgb)
2. unobserved confounding sensitivity analysis
   - Oster-style sensitivity curve
   - tipping-point heatmap